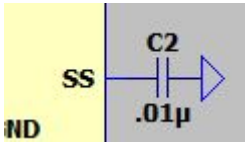
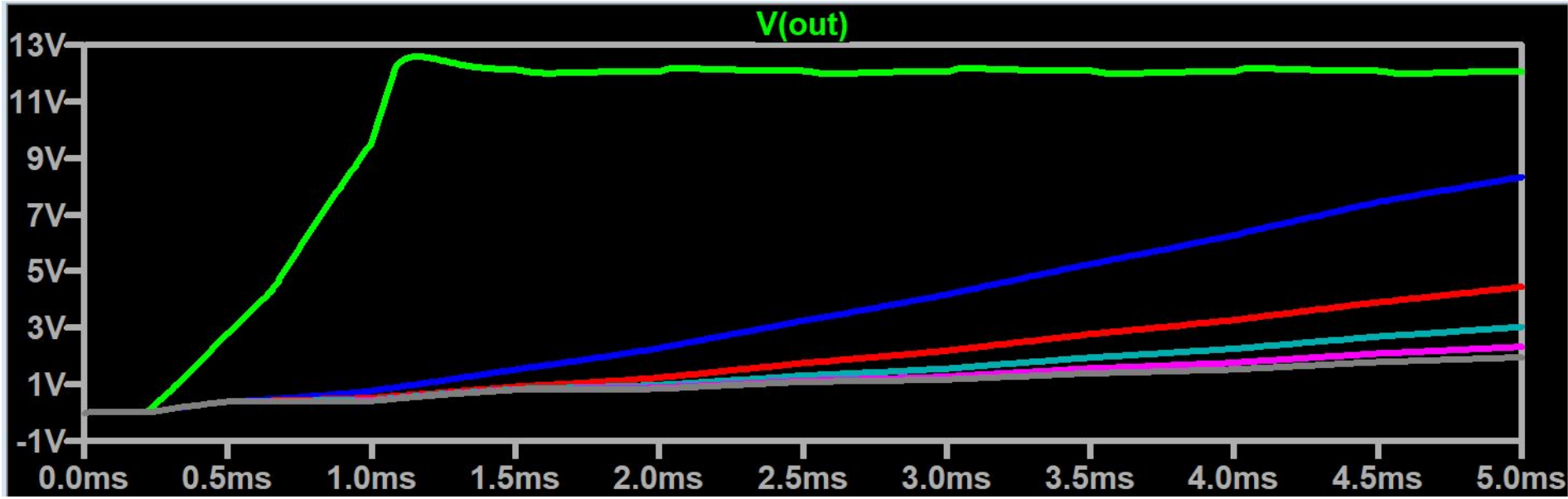
Phase 6 – Auxiliary Circuits

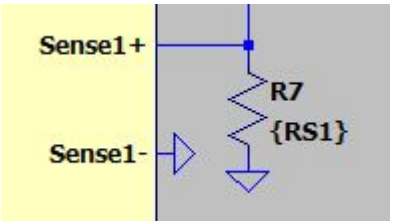
Soft Start Cap

.step param CSS 0.01u 0.5u 0.1u

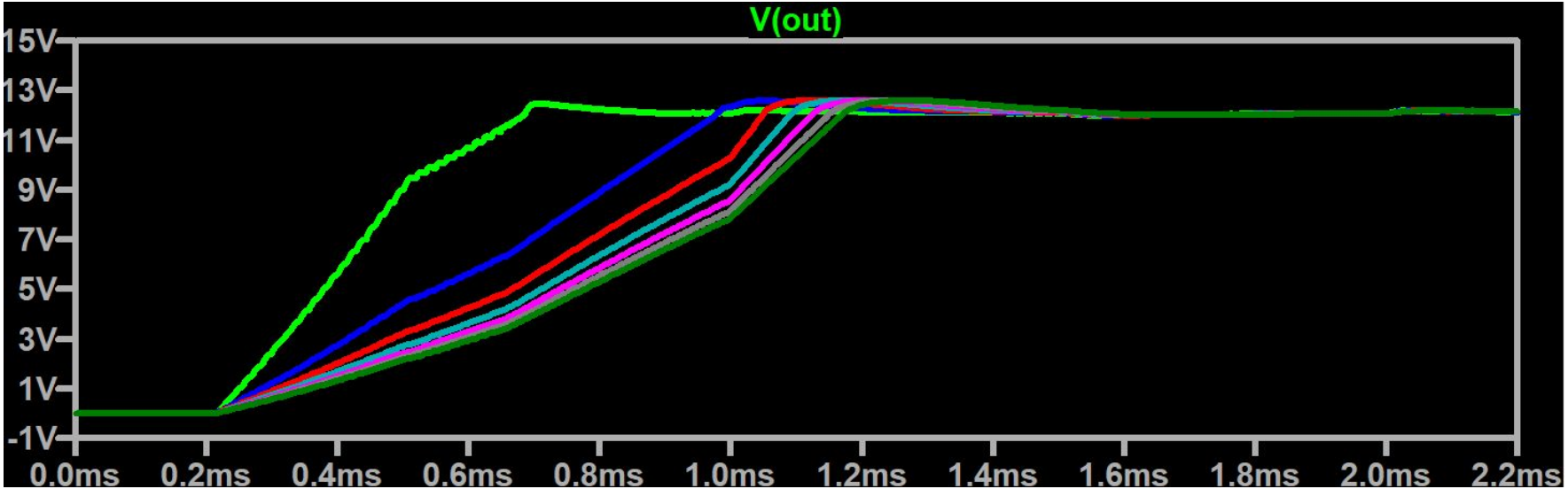




Current Sense 1



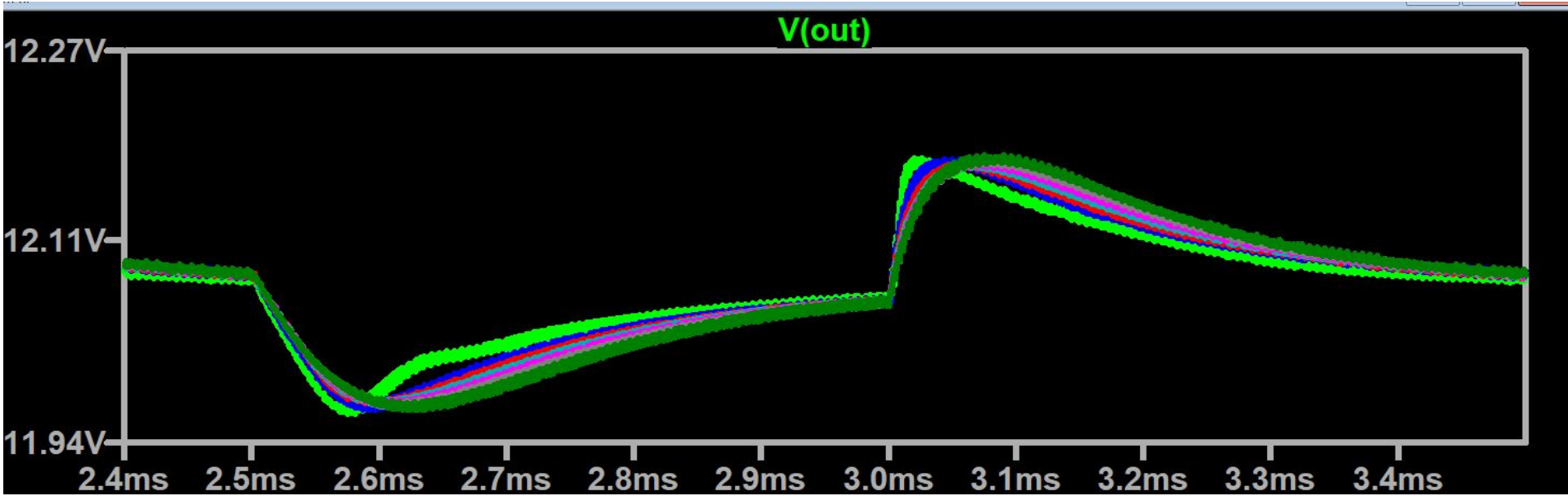
Transient Analysis



.step param RS1 1m 10m 1.5m

CS1

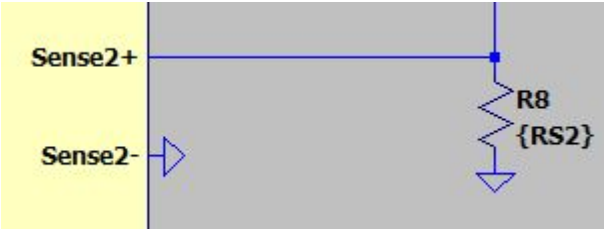
Steady-State

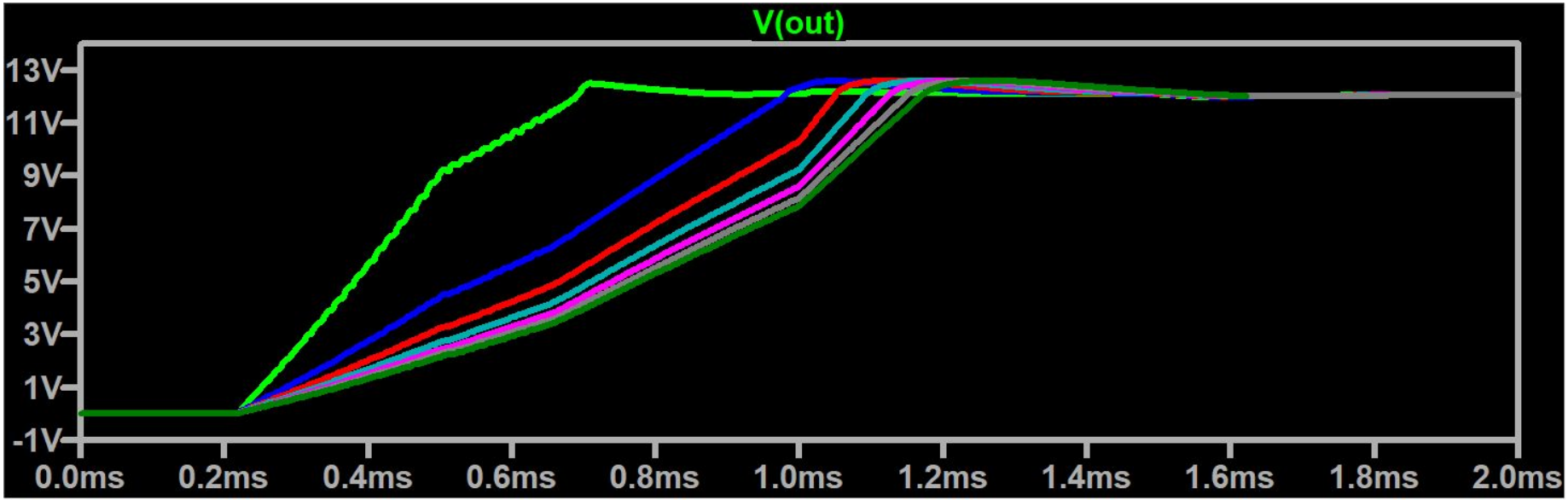


Current Sense 2

Transient Response

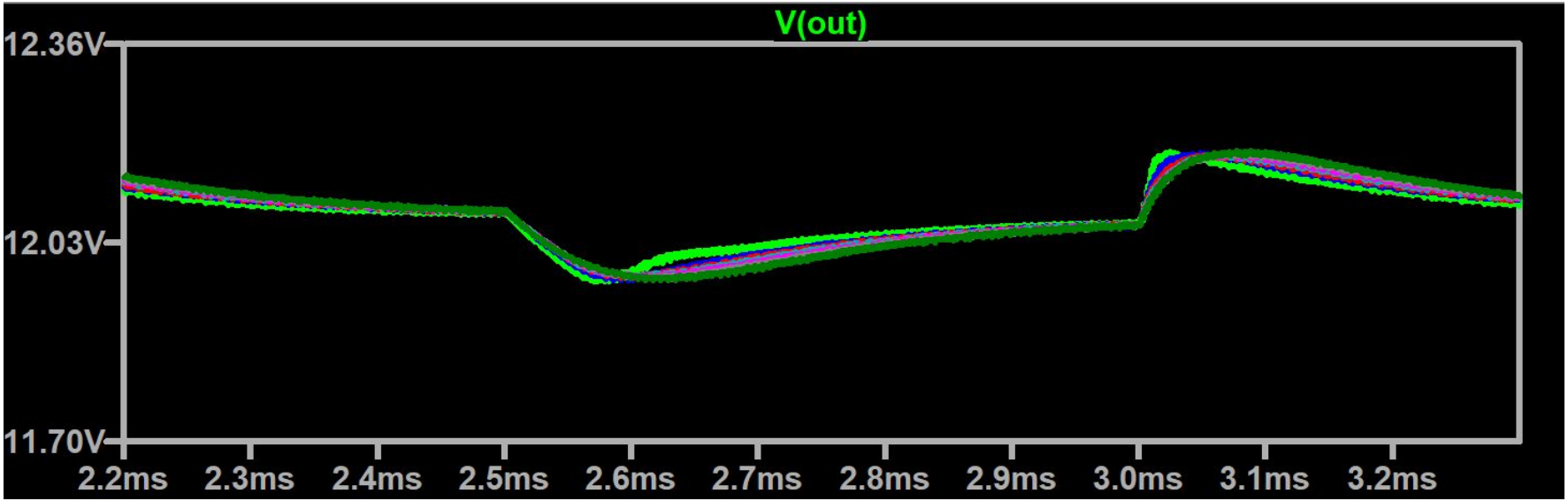
.step param RS2 1m 10m 1.5m





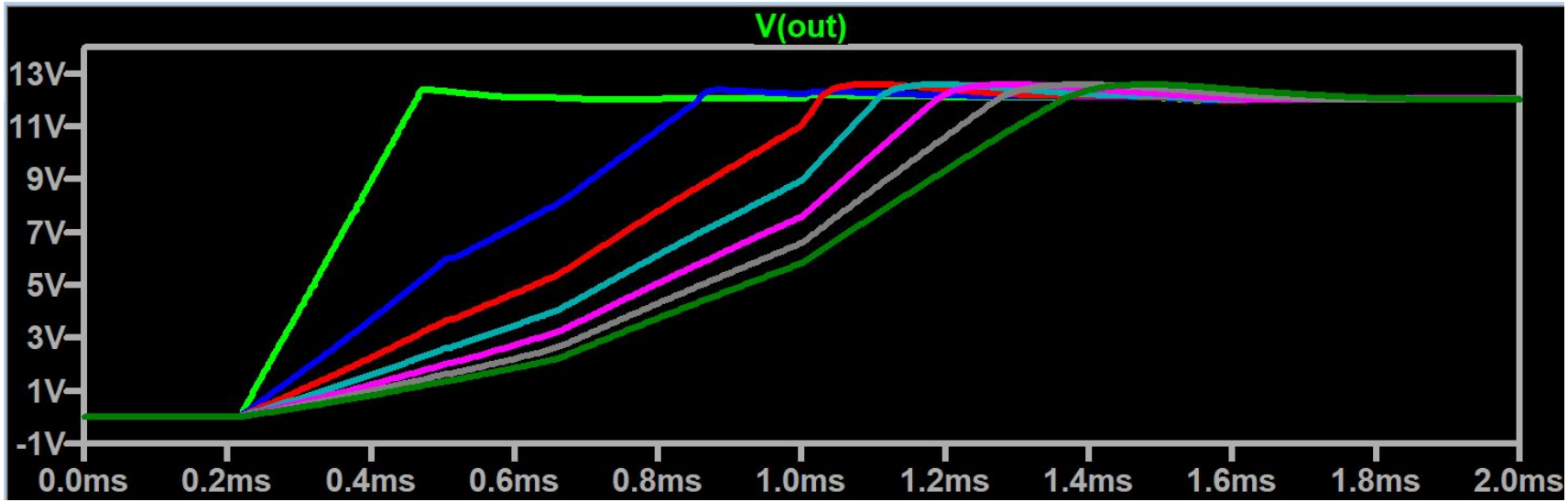
CS2

Steady-State



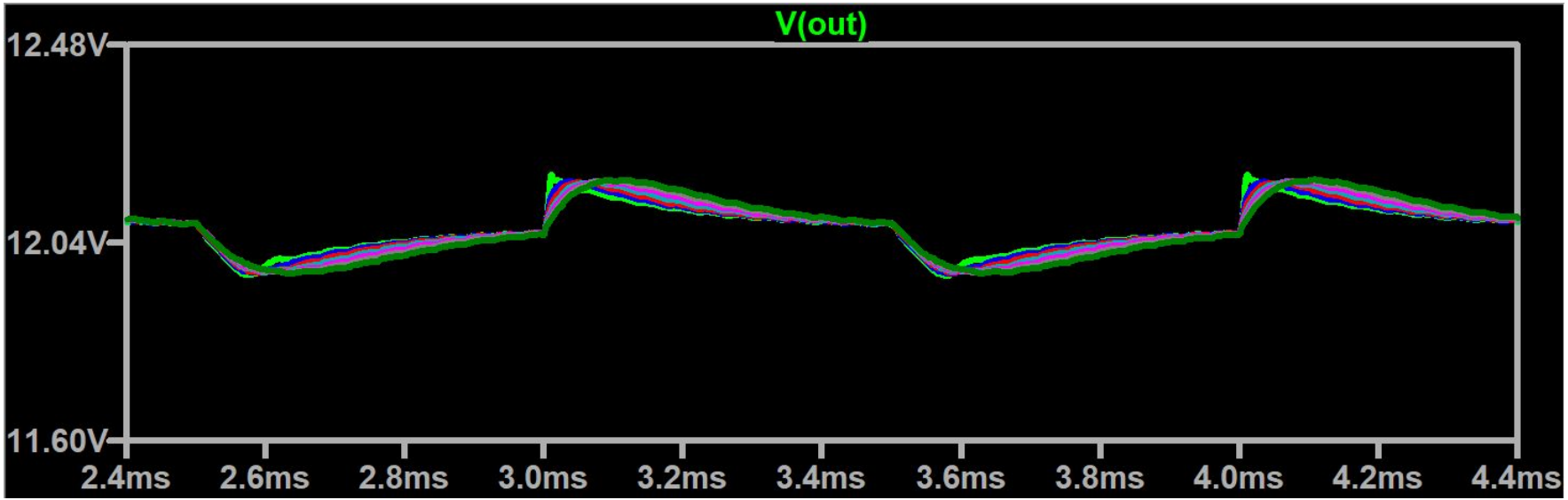
Changing both Sense 1+ and Sense 2+

Transient Response



Changing both Sense 1 and Sense 2

Steady State



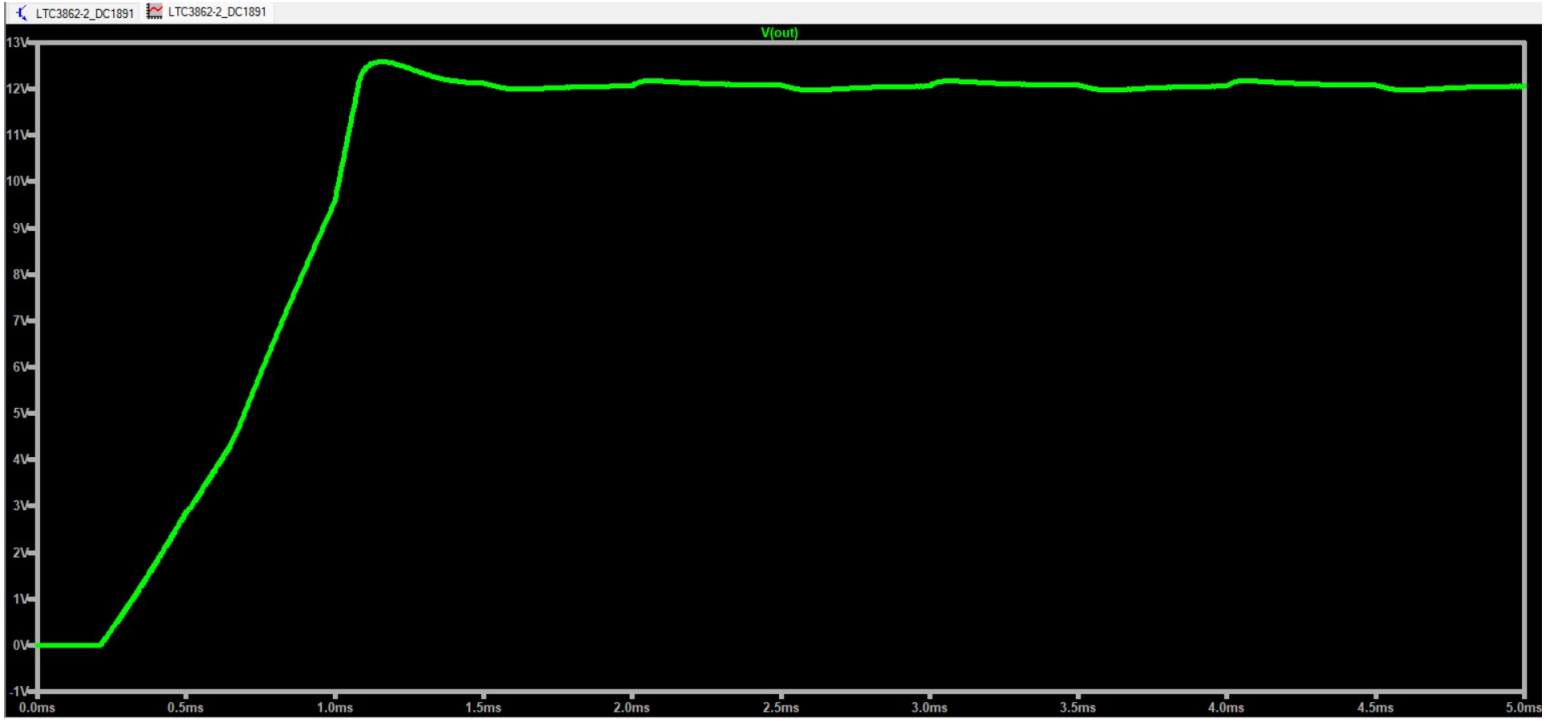
PHASE 7

- Complete simulations of auxiliary circuits.
- Note the relevant schematic details showing features/limits of operation for auxiliary circuits.
- Make slides detailing these auxiliary circuits and the supporting waveforms
- Do load step and line step simulations for your board.

Load Regulation

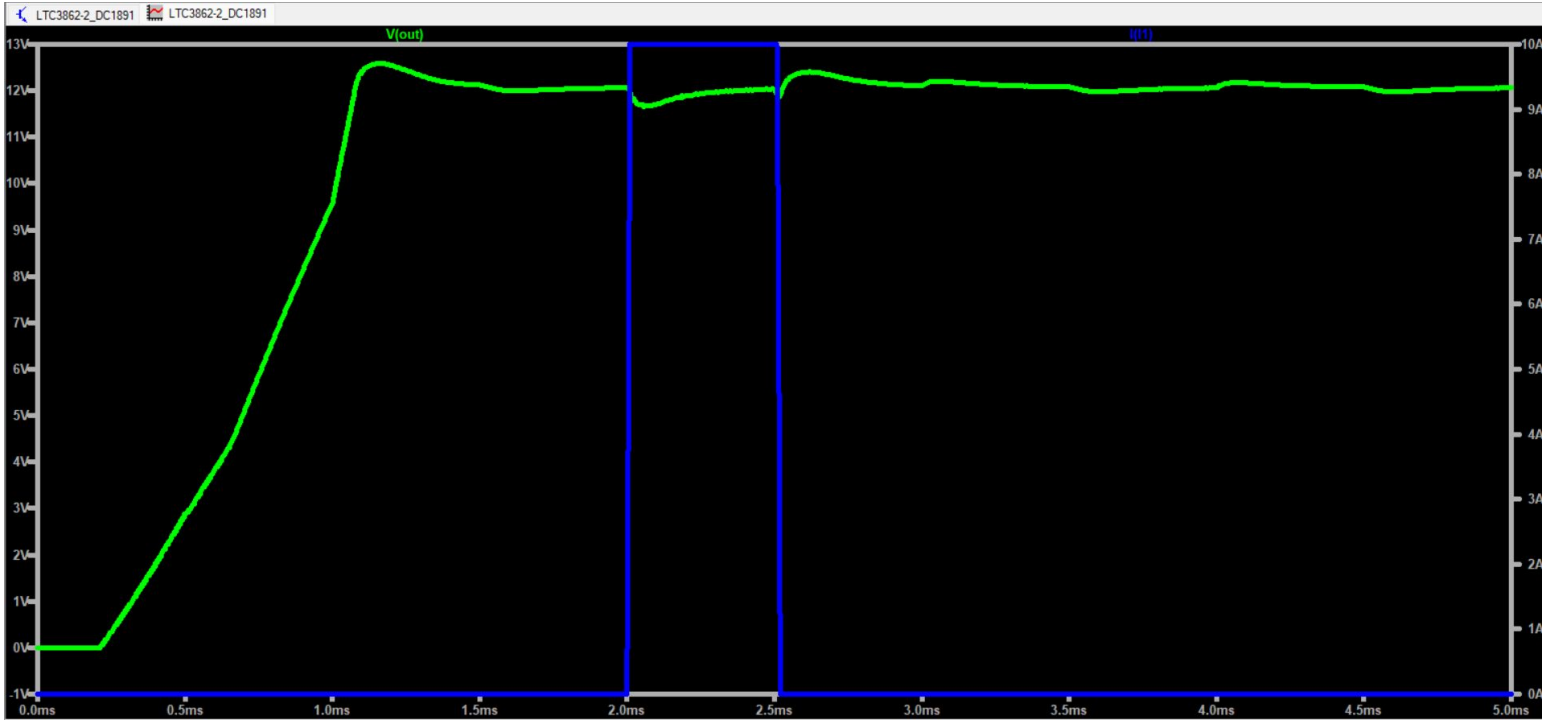
Our load regulation tests apply pulsed output current steps (0-10A and 0-20A for 0.5ms) while holding the input conditions constant. This shows how well the converter maintains the 12V setpoint under sudden changes in load demand.

Load Regulation - No Load Current Changes



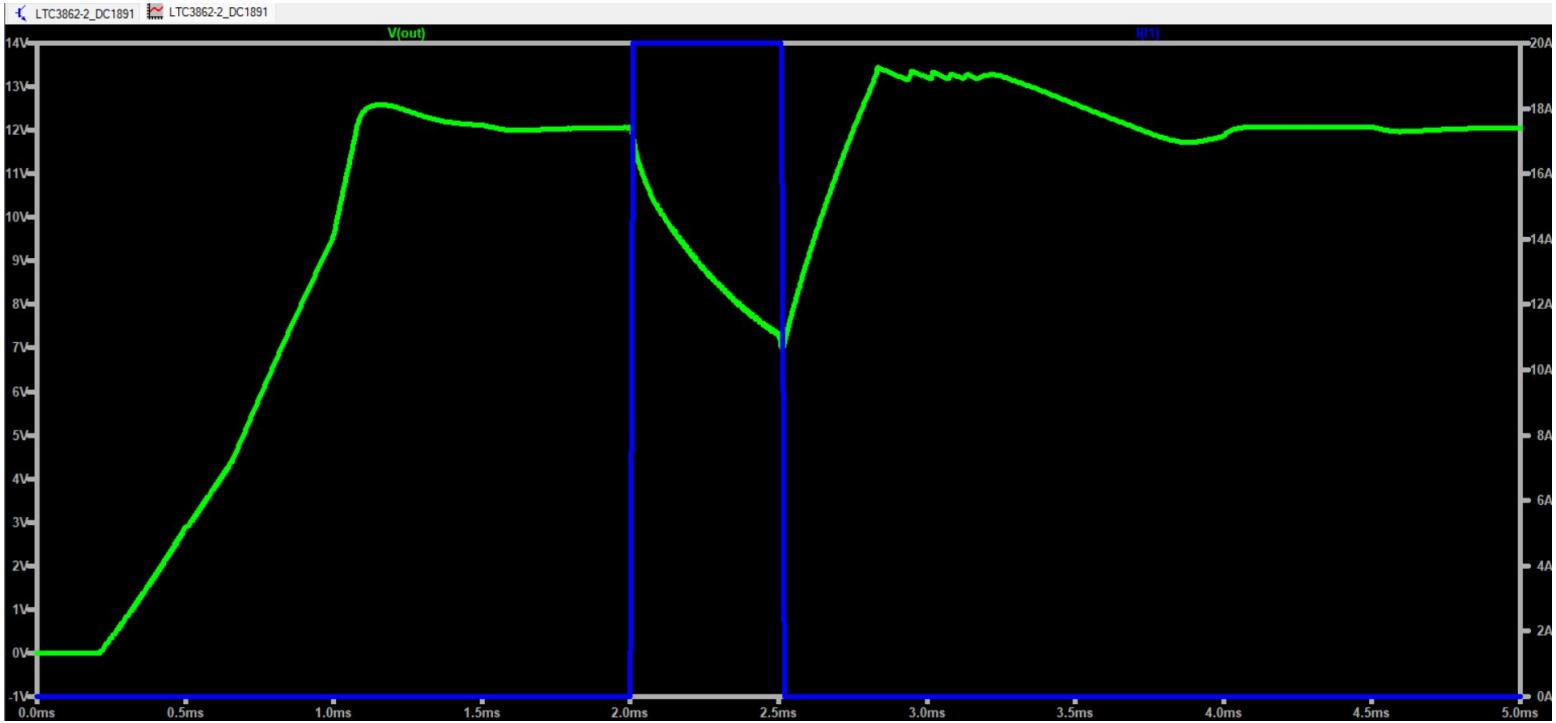
Load Regulation Test 1

With a current pulse of 0 to 10 A, rise and fall of 10us:



Load Regulation Test 2

With a current pulse of 0 to 20 A, rise and fall of 10us:



Load Regulation Analysis

- 0 to 10 A pulse (0.5 ms): Vout dips because the load jumps faster than the converter can ramp current, so Cout supplies the difference until the loop catches up.

- Return to 0 A: Vout overshoots because inductor energy + loop delay briefly over-deliver current, so Cout charges up before regulation backs off.

- 0 to 20 A pulse: the droop/overshoot are larger because the step is bigger, so the converter relies more on Cout and is more likely to run into current/duty limits, slowing correction.

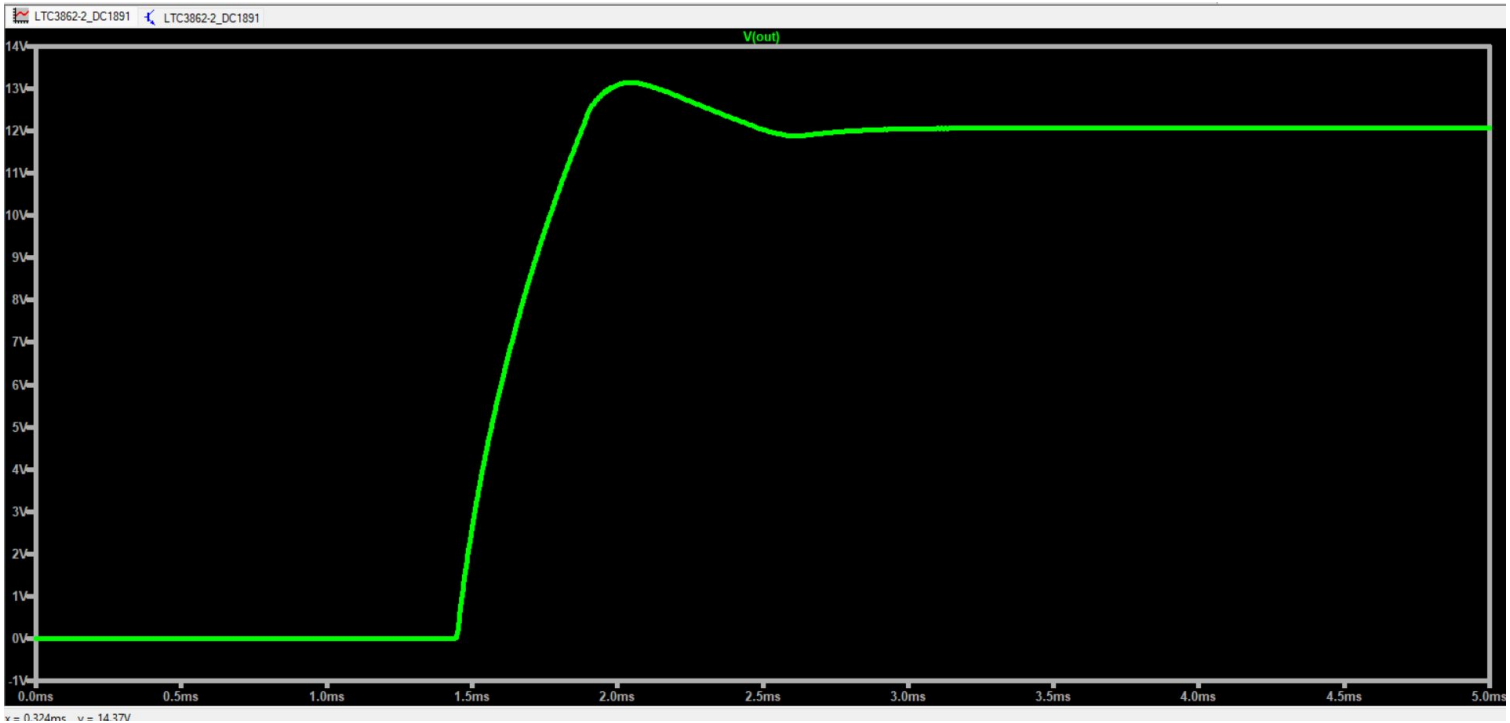
- Overall this behavior is expected and are normal load regulation effects from finite energy storage and finite control-loops.

Line Regulation

Our circuit by default had a pulsing input (0V to 14V) and static DC input (6V), so line regulation was being demonstrated from the get-go. We will conduct more line regulation tests with greater pulsing magnitude, and 2 pulsing inputs.

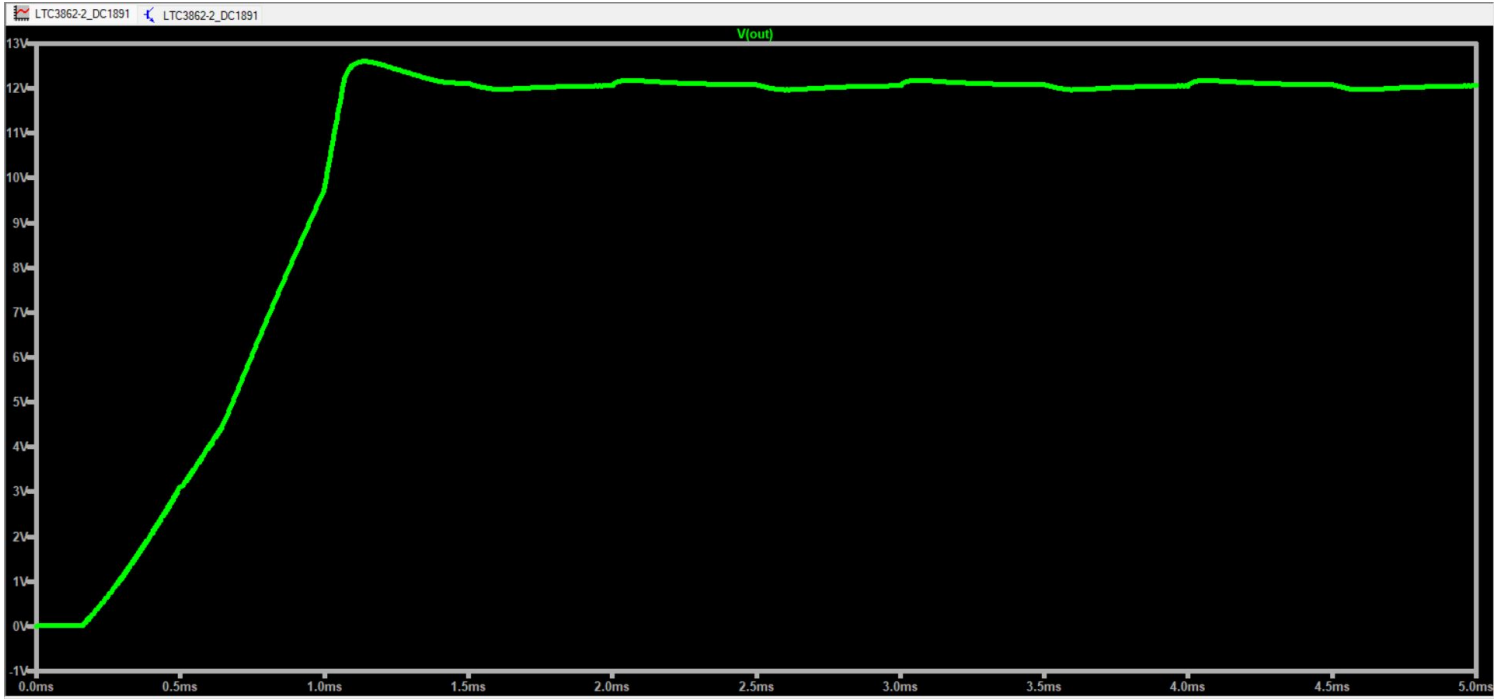
Line Regulation

6V DC input only



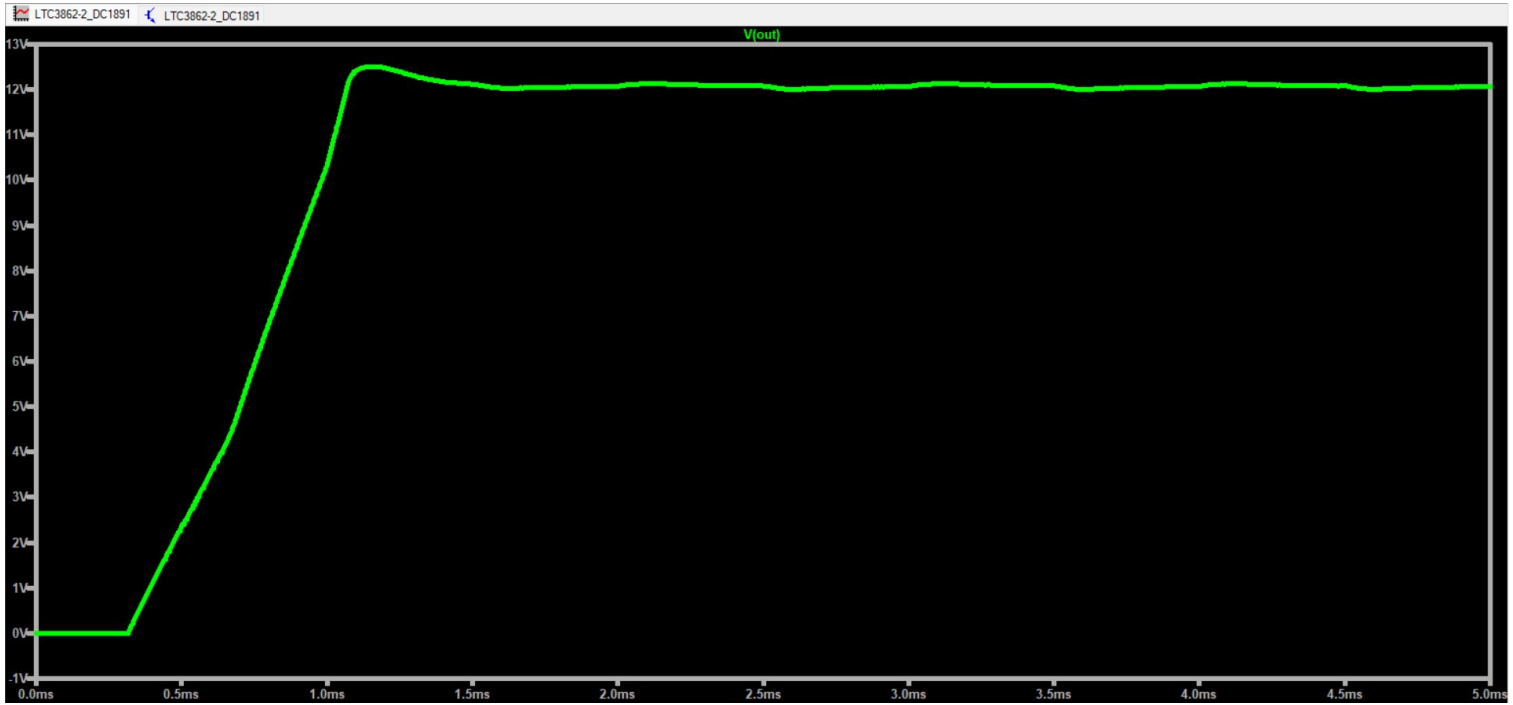
Line Regulation

6V DC input with 0 to 22V Pulse



Line Regulation

0V to 8V Pulse with 0V to 14V Pulse (offset)



Line Regulation Analysis

- The DC input acts as expected getting boosted up to 12V after a little oscillation and then settling to a very steady 12V.

- We see a small ripple for the 6V DC + 0-22V pulse, but still overall shows 12V DC at Vout.

- Surprisingly, there’s an even smaller ripple for the two pulsed inputs (0-8V + 0-14V).

- Overall, the line regulation of this converter is very good and regulates very well across a variety of different voltage inputs.